Modular Avionics Unit (MAU) (Mod 6/2049) – General Data

1. General

The Modular Avionics Unit (MAU), with the modules installed acts like the central processing unit for Central Computer. The MAU and modules, collect, compile and formats the data for the required system display. The Modular Avionics Unit (MAU) consists of a cabinet which houses a variety of line replaceable modules (LRUs).

2. Description

The MAU is installed on the upper of the three avionics racks between the nose baggage compartment and the flight compartment at approximately station 50. It consists of a cabinet/chassis containing a backplane circuit card assembly, two cooling fans and 12 user module slots that host a variety of line replaceable modules.

28 Volt DC power is provided from the left bus through 5 amp MAU 1A (E1) and MAU 2A (F1) left bus circuit breakers, and from the right bus through 5 amp MAU 1B (E2) and MAU 2B (F2) right bus circuit breakers on avionics circuit breaker panel (refer to 24-00-00, Electrical Power – General Data).

The MAU cabinet is divided into two channels (A and B), each channel is electrically isolated from the other with its own power supply module, Network Interface Controller (NIC) module and data communications backplane. The dual channel architecture of the MAU allows system functions to be distributed between channels. All modules are field replaceable and software updates can be loaded in the field. The modules communicate to the Avionics Standard Communications Bus (ASCB) and the Local Area Network (LAN) bus via the NIC modules.

The Avionics Standard Communications Bus (ASCB) consists of two independent busses, the No. 1 ASCB and No. 2 ASCB corresponding to pilot and co-pilot side primary data. Each NIC in the system reads and writes to the onside primary bus, and reads from the cross-side primary bus. One end of each ASCB is terminated within the wiring harness, the other end is terminated at a D series connector on the maintenance panel. When the maintenance panel connector is not being used, a terminating backshell must be installed. The ASCB bus uses special twisted pair wiring and is terminated at each point using an ASCB stub, refer to wiring manual PSM 1-64-2W for more details.

The LAN bus uses a 10base2 Ethernet communication scheme, and is used primarily for transferring data, such as MAU updates or changing MAU settings. It uses a triax cable which allows the outer shield to be terminated in multiple places, while the inner shield is only terminated at one end. Both ends of the LAN bus are available as BNC connectors on the maintenance panel. A 50 ohm BNC terminator must be connected to the BNCs when not in use. The LRUs connected to the LAN bus are connected in series; the bus enters the LRU on one pin and exits to the next LRU on another pin. Therefore, if the system is to be energized with any of the LRUs removed, the input and output pins must be jumpered.

The modules used in the Modular Avionics Unit include:

A. Two – Power Supply Modules (PS)

Each channel of the MAU has a dedicated, independent power supply. Either power supply can operate both of the MAU cabinet cooling fans. Each channel receives power from the left and right bus. The PS modules contain no processing or backplane communication capability.

B. Two – Network Interface Controller (NIC) Modules

The Network Interface Controller (NIC) modules provide a gateway for the MAU modules to access the ASCB and LAN bus. Two NIC modules are installed, one for each channel of the MAU.

Two Aircraft Personality Modules (APMs) are located in the MAU NIC backshells. They are configuration modules which contain Apex configuration data, such as system identifier, aircraft type, aircraft serial number, installed configuration options and system settings. The APM allows individual cards within the MAU to be removed and replaced quickly, without the need to reconfigure the card for a specific aircraft.

1) The NICs interface to the following aircraft systems:

   a) ID straps

   b) ASCB 1 & 2

   c) APMs

   d) LAN buses

C. Two – Advanced Graphics Module (AGM)

The Advanced Graphics Module (AGM) is a single channel module, one per MAU channel is installed. The AGM does general purpose processing as well as display processing and graphics generation. The Configuration Management Function (CMS), Flight Management System (FMS), electronic display of Jeppesen charts function and maintenance functions (CMC, ACMS) are also hosted on the AGM module. AGM 1 (MAU channel A) drives the pilot's PFD and upper MFD and AGM 2 (MAU channel B) drives the co-pilot's PFD and lower MFD.

A repeater capability allows the pilot's PFD to be displayed on the co-pilot's PFD (and vice-versa) in the event of a single AGM failure. The PFD controllers, MF controller and reversion panel are interfaced with the AGMs. AGM integrity is monitored by the Monitor Warning Function (MWF), which verifies that the data selected by the AGM for display on the DUs has high integrity.

1) The AGMs receive inputs from the following aircraft systems:

   a) Ethernet

   b) PFD Controllers

   c) MF Controller

   d) Reversion Panel

   e) EGPWS

   f) Weather Radar

   g) XM Weather (if installed)

   h) Stormscope (if installed)

2) The AGMs output to the following aircraft systems:

   a) Display Units

   b) Stormscope (if installed)

D. Custom Input/Output (CSIO) Modules

The Custom I/O (CSIO) Module is a dual channel module. Each channel is connected to a different MAU backplane (channel A and B). The CSIO module translates aircraft model-specific I/O data onto and off of the ASCB. The CSIO also translates selected fuselage signals from analog to digital, and then sends this digital data to the Data Acquisition Unit (DAU).

1) The CSIO module received inputs from the following aircraft systems:

   a) DMEs

   b) DAUs

c) Audio Panels

d) RAD ALT

e) EGPWS

f) FDR Accelerometer

g) XPDRs

h) GPSs

i) TCAS

j) Aircraft Voltages

2) The CSIO module outputs to the following aircraft systems:

a) DAUs

b) Audio Panels

c) FDR (if installed)

E. Two – Actuator Input/Output and Processor (AIOP) Modules

The Actuator I/O Processor (AIOP) Module is a single channel module, one per MAU channel is installed. The AIOP module is principally associated with the Automatic Flight Control System (AFCS) and flight controller operation. Currently this functionality is limited to selected minimums, heading or track, and selected altitude.

1) The AIOP modules receive inputs from the following aircraft systems:

a) Flight Controller

b) ADAHRS

2) The AIOP modules output to the following aircraft systems:

a) Flight Controller

F. Generic Input/Output (GIO) Module

The Generic Input/Output (GIO) Module is a dual channel module. Each module channel is connected to a different MAU backplane (channel A and B). The GIO module translates aircraft I/O data onto and off of the ASCB via the MAU's backplane. The GIO module also translates selected fuselage discretes for ARINC 429, which connects to the Data Acquisition Unit (DAU).

1) The GIO module receives inputs from the following aircraft systems:

a) ADAHRS

b) FDR (if installed)

c) CVR (if installed)

d) EGPWS

e) DAUs

f) Master Warning/Caution

g) XPDRs

h) GPSs

i) TCAS

j) Display Units

k) Aural Alert Power (mute)

l) Caution Dim Box (lamp test)

m) Various Aircraft Discrete Signals

2) The GIO module outputs to the following aircraft systems:

a) ADAHRS

b) CVR (if installed)

c) EGPWS

d) MMDRs

e) Master Warning/Caution

f) MF Controller

g) XPDRs

h) GPSs

i) TCAS

j) ELT

k) Weather Radar

3. Adjustment/Test

A. Modular Avionics Unit (MAU) Test

The MAU has extensive built in test and fault detection and exclusion capability. Failures of components or loss of system capability will be announced to the crew using the aircraft crew alerting system. Information may be presented as caution, advisory, or status messages, depending on the importance of the message or the severity of the condition.

1) Ensure MAU 1A (E1), MAU 2A (F1), MAU 1B (E2) and MAU 2B (F2) circuit breakers on the avionics circuit breaker panel are closed.

2) Ensure none of the following CAS messages are present following system initialization:

a) MAU Fail

b) ASCB Fail

c) AIOP Fail

d) CSIO Fail

e) MAU Overheat

f) MAU FAN Fail

g) GIO Fail

h) AGM Fail

i) CMS Fail

j) FMS Fail

3) Ensure the following windows are displayed properly:

a) Altitude Director Indicator (ADI)

b) Airspeed

c) Altitude

d) Vertical Speed

e) Horizontal Situation (HSI) Displays and Annunciators

f) Radio Management

g) Situational Awareness and Flight Planning

B. Modular Avionics Unit (MAU) Software Load or Update

For the procedure for loading the APEX MAU Software refer to 45-00-02, Apex Modular Avionics Unit Software – Load Procedures.

C. Options File or Setting File Load or Update

For the procedure for loading the APEX Options or Setting File refer to 45-00-01, Apex Options and Settings File – Load Procedures.

D. FMS Performance Database File Load or Update

For the procedure for loading the APEX FMS Aircraft Performance Database File refer to 45-00-03, Apex FMS Aircraft Performance Database Files – Load Procedures.

E. Ferrite Bead Verification

1) Remove NIC 1 or 2 backshell. Refer to Para 4.C.

2) Remove fasteners from backshell cover and remove cover.

3) Verify presence of ferrite beads on LAN wires. The LAN wires are the coaxial cables terminated to the center post of the backshell.

4) If ferrite beads are not present, install according to PSM 1-64-2W, Chapter 20 for details.

5) Install cover onto NIC backshell and tighten fasteners.

6) Install NIC 1 or 2 backshell. Refer to Para 4.D.

4. Removal/Installation

WARNING **INSTALLATION OF THIS COMPONENT CONTAINS PROTECTION FEATURES THAT ARE CRITICAL TO PREVENTING DEGRADATION FROM THE EFFECTS OF HIGH INTENSITY RADIATED FIELDS (HIRF) AND INDIRECT EFFECTS OF LIGHTNING (IEL). ENSURE INTEGRITY OF THE HIRF/IEL PROTECTION FEATURES ARE MAINTAINED IN ACCORDANCE WITH THE FOLLOWING:**

CHASSIS GROUNDING – ALL RACK MOUNT/REMOTE MOUNT UNITS SHALL BE ELECTRICALLY BONDED TO THE AIRFRAME. ENSURE THAT THE MATING SURFACES BETWEEN THE LRU MOUNTING TRAY (OR LRU MOUNTING FEET IF A TRAY IS NOT USED) PROVIDE A LOW IMPEDANCE ELECTRICAL PATH. ALL MATING SURFACES MUST BE FREE OF ALL PAINT AND OTHER NONCONDUCTIVE ELEMENTS. REFER TO BONDING REQUIREMENTS IN 20-75-00, ELECTRICAL BONDING AND GROUNDING OF AIRCRAFT STRUCTURES – GENERAL DATA.

SHIELD GROUNDING – VERIFY SHIELD GROUNDING AT CONNECTOR BACKSHELLS AS PER REQUIREMENTS IN PSM 1-64-2W.

BACKSHELLS OR STRAIN RELIEFS – VERIFY PROPER ELECTRICAL BONDING BETWEEN THE BACKSHELL/STRAIN RELIEF AND THE MOUNTING TRAY OR LRU CASE AS PER REQUIREMENTS IN PSM 1-64-2W.

CAUTION **THE MAU AND MODULES ARE ELECTROSTATICALLY DISCHARGE SENSITIVE (ESDS). ENSURE PROPER ESDS PROCEDURES ARE FOLLOWED. FAILURE TO DO SO COULD DAMAGE UNIT.**

CAUTION **THE MAU AND COMPONENTS ARE ESDS SENSITIVE AND ANY WORK DONE ON THE MAU AND COMPONENTS AWAY FROM AIRCRAFT MUST BE DONE ON AN ESDS-PROTECTED WORK AREA. FAILURE TO DO SO MAY DAMAGE UNIT.**

CAUTION **ENSURE AN ESDS WRIST STRAP IS WORN AND IS CONNECTED TO THE GROUND LOCATED ON THE MAU BEFORE DOING THE REMOVAL OR INSTALLATION OF THE MAU.**

CAUTION **CARE MUST BE TAKEN NOT TO DAMAGE THE AIR OUTLETS SCREENS ON THE REAR OF THE MAU WHEN YOU REMOVE OR INSTALL UNIT.**

CAUTION **DAMAGE TO MAU AND COMPONENTS OR DEGRADATION OF PERFORMANCE CAN OCCUR IF THE CONNECTOR PINS OF THE MAU OR COMPONENTS ARE TOUCHED.**

A. Removal of the MAU

1) Ensure the aircraft electrical system is OFF.

2) Pull circuit breaker MAU 1A (E1), MAU 2A (F1) (Left Bus), MAU 1B (E2), MAU 2B (F2) (Right Bus), on grid 1 and 2 of the avionics circuit breaker panel and fit safety clips.

3) Gain access to the forward avionic shelves through the forward baggage compartment.

4) Remove top cover attached to bulkhead station 37.21.

5) Remove the MAU backshells, refer to Para 4.C.

6) Remove the (2) cable clamps and reposition cable harness to allow greater access.

7) Loosen the (4) retaining screw clamps which secure MAU to mounting tray.

8) Carefully move MAU forward until unit is clear of mounting wedge located at the rear of the mounting tray.

Note The MAU, with all the components installed, weighs approximately 40 lbs, personnel must take care in lifting the unit.

9) Lift MAU off avionics shelf and clear of aircraft.

10) Visually inspect the metal mesh covering the air intakes for dust or debris. Clean if required.

CAUTION DO NOT BLOW DUST OR DEBRIS INTO MAU COVER.

B. Installation of MAU

1) Visually inspect the metal mesh covering the air intakes for dust or debris. Clean if required.

CAUTION DO NOT BLOW DUST OR DEBRIS INTO MAU COVER.

2) Ensure the aircraft electrical system is OFF.

3) Ensure the MAU 1A (E1), MAU 2A (F1) (Left Bus), MAU 1B (E2), MAU 2B (F2) (Right Bus) on grid 1 and grid 2 of the avionics circuit breaker is pulled and fitted with safety clips.

4) Gain access to the forward avionic shelves through the forward baggage compartment.

5) Remove top cover attached to bulkhead station 37.21.

6) Verify that the static bonding points to the aircraft frame are clean.

7) If you are installing a different MAU into aircraft, make sure all 12 modules are installed correctly and are secure within the MAU cabinet, refer to Para 4.E and Para 4.F.

8) Inspect the condition of the MAU support bracket. If the tray is damaged, remove and replace as per Para 4.I and Para 4.J.

Note The MAU, with all the components installed, weighs approximately 40 lbs, personnel must take care in lifting the unit.

9) Lift MAU into the forward avionics compartment and onto the mounting tray.

10) Carefully move the MAU aft in the tray until rear of unit sits securely in the mounting wedge at the rear of the mounting tray.

11) Tighten the (4) retaining screw clamps, torque to 42 ± 2 in–lbs, ensure MAU is secure.

12) Reposition MAU cable harness and install (2) cable clamps.

13) Install MAU backshells, refer to Para 4.D.

14) Ensure all connections are secure.

15) Ensure proper bonding has been achieved between the MAU and the airframe.

16) Install top cover attached to bulkhead station 37.21. Ensure cover is secure.

WARNING ACCESS COVER IS REQUIRED TO MAINTAIN PROTECTION FEATURES THAT ARE ESSENTIAL TO MINIMIZING ADVERSE EFFECTS FROM HIGH INTENSITY RADIATED FIELDS (HIRF).

Note If new MAU is being installed or has one or more modules installed, MAU software will have to be loaded. (Refer to Chapter 45 for loading procedures).

17) Remove safety clips and close MAU 1A (E1), MAU 1B (E2), MAU 2A (F1) and MAU 2B (F2) circuit breakers on the avionics circuit breaker panel.

18) Do an operational test on the MAU, refer to Para 3.A.

C. Removal of MAU Backshell

CAUTION **ENSURE AN ESDS WRIST STRAP IS WORN AND IS CONNECTED TO THE GROUND LOCATED ON THE MAU BEFORE DOING THE REMOVAL OR INSTALLATION OF THE MAU BACKSHELLS.**

CAUTION **DAMAGE TO MAU AND COMPONENTS OR DEGRADATION OF PERFORMANCE CAN OCCUR IF THE CONNECTOR PINS OF THE MAU OR COMPONENTS ARE TOUCHED.**

These removal and installation procedures are applicable for all the backshells installed in the MAU Cabinet.

1) Ensure the aircraft electrical system is OFF.

2) Open MAU 1A (E1), MAU 2A (F1), MAU 1B (E2) and MAU 2B (F2) circuit breakers on the avionics circuit breaker panel and fit safety clips.

3) Gain access to the forward avionics shelf through the forward baggage compartment.

4) Remove top cover attached to bulkhead station 37.21.

5) Loosen the top bolt from the backshell until it has disengaged from MAU.

6) Carefully tilt the backshell away from the MAU until the alignment pin is clear of the alignment hole, approximately mid way on the module.

7) Disengage the bottom backshell hook from the base of the MAU and move backshell away from unit.

8) Install protective covers on backshell.

D. Installation of MAU Backshell

CAUTION **DAMAGE TO THE MODULE CONTACTS MAY RESULT IF THE FOLLOWING PROCEDURES ARE NOT CARRIED OUT WITH PROPER CARE AND ATTENTION.**

1) Ensure the aircraft electrical system is OFF.

2) Ensure MAU 1A (E1), MAU 2A (F1), MAU 1B (E2) and MAU 2B (F2) circuit breakers on the avionics circuit breaker panel are open and fitted with safety clips.

3) Remove protective cover from backshell.

4) Place the backshell to engage hook onto the base of the MAU.

5) Carefully align the pin into the alignment hole in the module.

6) Install the top bolt into MAU and torque to 32 in–lbs. Ensure the backshell is secure.

7) Install top cover attached to bulk head station 37.21. Ensure cover is secure.

8) Remove safety clips and close MAU 1A (E1), MAU 2A (F1), MAU 1B (E2) and MAU 2B (F2) circuit breakers on the avionics circuit breaker panel.

9) Do an operational test on the MAU, refer to Para 3.A.

E. Removal of Modules

CAUTION **ENSURE AN ESDS WRIST STRAP IS WORN AND IS CONNECTED TO THE GROUND LOCATED ON THE MAU BEFORE DOING THE REMOVAL OR INSTALLATION OF THE MAU MODULES.**

CAUTION **DAMAGE TO MAU AND COMPONENTS OR DEGRADATION OF PERFORMANCE CAN OCCUR IF THE CONNECTOR PINS OF THE MAU OR COMPONENTS ARE TOUCHED.**

These removal and installation procedures are for all the modules installed in the MAU Cabinet. Please note the individual modules may have two or four D connectors securing module into the MAU Cabinet.

1) Remove backshell from module as per Para 4.C.

2) Install protective covers on the module connectors.

CAUTION **DAMAGE TO THE MODULE OR MAU MAY RESULT IF THE FOLLOWING PROCEDURES ARE NOT CARRIED OUT WITH PROPER CARE AND ATTENTION.**

3) Loosen the jackscrews gradually and evenly until they are all free from the MAU threads.

4) While gripping the jackscrews, carefully pull the module clear of the MAU cabin slot.

5) Place module into ESDS approved protective sleeve.

F. Installation of Modules

1) Ensure the aircraft electrical system is OFF.

2) Ensure MAU 1A (E1), MAU 2A (F1), MAU 1B (E2) and MAU 2B (F2) circuit breakers on the avionics circuit breaker panel are open and fitted with safety clips.

3) Remove module from ESDS approved protective sleeve.

4) Carefully align module into applicable slot.

5) Gently push module into slot, ensure module is all the way in.

6) Tighten the jackscrews gradually and evenly to 42 in–lbs. Ensure the module is secure.

7) Remove protective covers from module connectors.

8) Install backshell, refer to Para 4.D.

9) Remove safety clips and close MAU 1A (E1), MAU 2A (F1), MAU 1B (E2) and MAU 2B (F2) circuit breakers on the avionics circuit breaker panel.

Note If a new module has been installed, MAU software will have to be loaded. (Refer to Chapter 45 for loading procedures).

10) Do an operational test on the MAU, refer to Para 3.A.

G. Removal of MAU Cooling Fans

CAUTION **ENSURE AN ESDS WRIST STRAP IS WORN AND IS CONNECTED TO THE GROUND LOCATED ON THE MAU BEFORE DOING THE REMOVAL OR INSTALLATION OF THE COOLING FANS FROM THE MAU.**

CAUTION **CARE MUST BE TAKEN NOT TO DAMAGE THE AIR OUTLET SCREENS ON THE REAR OF THE MAU WHEN YOU REMOVE OR INSTALL A COOLING FAN.**

CAUTION **DAMAGE TO MAU AND COMPONENTS OR DEGRADATION OF PERFORMANCE CAN OCCUR IF THE CONNECTOR PINS OF THE MAU OR COMPONENTS ARE TOUCHED.**

These removal and installation procedures are applicable for the cooling fans located in the rear of the MAU.

1) Remove MAU as per Para 4.A.

2) Unlatch the (3) clamps which hold the rear fan hinge plate, located at the rear of the unit.

3) Pull the fan out of the slot, towards the top of the hinge. This disconnects the electrical connector on the fan, so some resistance may be felt.

4) Put fan in an ESDS approved protective sleeve.

5) Visually inspect the metal mesh covering the air intakes for dust or debris. Clean as required.

CAUTION **DO NOT BLOW DUST OR DEBRIS INTO THE INTERIOR OF MAU.**

H. Installation of MAU Cooling Fans

1) Visually inspect the metal mesh covering the air intakes for dust or debris. Clean if required.

CAUTION **DO NOT BLOW DUST OR DEBRIS INTO THE INTERIOR OF MAU.**

2) Remove fan from ESDS approved protective sleeve.

3) Slide fan into position on the fan hinge plate. Align the guide pin for the electrical connector and push the fan down until it is fully seated.

4) Close the fan hinge plate and secure with clamps.

5) Install MAU, as per Para 4.B.

6) Apply power to aircraft.

7) Ensure there is no “Mau Fan Failure Message” displayed through the CAS.

I. Removal of MAU Support Bracket

1) Remove the MAU as per Para 4.A.

2) Remove the (16) mounting screws which secure the bracket to the avionics shelf.

3) Remove bracket.

J. Installation of MAU Support Bracket

1) Visually inspect the mating surfaces of the tray and the avionics shelf for any corrosion or contaminants.

2) Ensure the mounting bracket is oriented correctly, mounting edge to the aft.

3) Install the (16) mounting screws which secure the bracket to the avionics shelf.

4) Ensure proper bonding has been achieved between the tray and the avionics shelf.

5) Install MAU as per Para 4.B.

5. Troubleshooting

A. Modular Avionics Unit (MAU) Software

1) Ensure MAU 1A (E1), MAU 2A (F1), MAU 1B (E2) and MAU 2B (F2) circuit breakers are closed.

If problem persists reboot MAU as follows:

a) Open MAU 1A (E1), MAU 2A (F1), MAU 1B (E2) and MAU 2B (F2) circuit breakers on the avionics circuit breaker panel.

b) Wait 60 seconds.

c) Close MAU 1A (E1), MAU 2A (F1), MAU 1B (E2) and MAU 2B (F2) circuit breakers on the avionics circuit breaker panel.

If problem persists:

d) Ensure the correct software is loaded by checking the systems window of the MFD.

e) Follow load procedures described in Chapter 45 if software is suspected.

B. MAU Connectors

1) Ensure both ASCB1 and ASCB2 dummy plugs are installed and seated correctly on the maintenance panel below the top FWD avionics shelf.

2) Ensure both LAN Bus Terminators are installed correctly on the maintenance panel.

If problem persists:

3) Open MAU 1A (E1), MAU 2A (F1), MAU 1B (E2) and MAU 2B (F2) circuit breakers on the avionics circuit breaker panel.

4) Ensure each connector on the MAU is fully seated and tight.

5) If found not seated correctly, remove the electrical connector that is suspected and visually check for damaged or pushed back contacts or damaged wiring. Refer to Para 4.C.

CAUTION **DO NOT TOUCH MAU MODULE CONTACTS. PROPER ESDS PROCEDURES MUST BE FOLLOWED. AN ESDS WRIST STRAP MUST BE WORN AND GROUNDED AT ALL TIMES, WHEN HANDLING THE MAU EQUIPMENT.**

6) Ensure the MAU module connector contacts are not bent or damaged.

7) Install the connectors which had been removed. Refer to Para 4.D.

8) Close MAU 1A (E1), MAU 2A (F1), MAU 1B (E2) and MAU 2B (F2) circuit breakers on the avionics circuit breaker panel.

9) Check if the problem has been resolved.

C. MAU System Wiring

1) Open MAU 1A (E1), MAU 2A (F1), MAU 1B (E2) and MAU 2B (F2) circuit breakers on the avionics circuit breaker panel.

2) If troubleshooting at a wire level is required, refer to the applicable wiring diagrams in PSM 1-64-2W.

D. MAU Hardware

1) If the problem can be isolated to one of the MAU modules the module should be replaced. Refer to Para 4.E and Para 4.F.

2) If further repair or service to the unit is required, send to appropriately rated avionics repair facility.

6. Servicing

A. Air Intake Screen

The metal mesh at the air inlet at the back of the MAU cabinet should be clean and free of lint, dust and other foreign objects which may block cooling air flow.

The metal mesh must be inspected for cleanliness after any of the following events:

1) MAU removal.

2) Servicing of the MAU cooling fans.